

Machine Learning Guided AQFEP: A Fast and Efficient Absolute Free Energy Perturbation Solution for Virtual Screening

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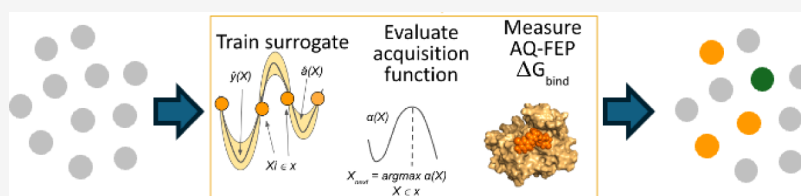


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ABSTRACT: Structure-based methods in drug discovery have become an integral part of the modern drug discovery process. The power of virtual screening lies in its ability to rapidly and cost-effectively explore enormous chemical spaces to select promising ligands for further experimental investigation. Relative free energy perturbation (RFEP) and similar methods are the gold standard for binding affinity prediction in drug discovery hit-to-lead and lead optimization phases, but have high computational cost and the requirement of a structural analog with a known activity. Without a reference molecule requirement, absolute FEP (AFEP) has, in theory, better accuracy for hit ID, but in practice, the slow throughput is not compatible with VS, where fast docking and unreliable scoring functions are still the standard. Here, we present an integrated workflow to virtually screen large and diverse chemical libraries efficiently, combining active learning with a physics-based scoring function based on a fast absolute free energy perturbation method. We validated the performance of the approach in the ranking of structurally related ligands, virtual screening hit rate enrichment, and active learning chemical space exploration; disclosing the largest reported collection of free energy simulations to date.

INTRODUCTION

In the past decade, the impact of structure-based methods in drug discovery has increased thanks to improvements in the accuracy of force fields, GPU computing, and machine learning.¹ A prime example of these advances is the development and implementation of structure-based virtual screening (VS) methods, which have become an integral part of the modern drug discovery process. The effectiveness of VS lies in its capacity to quickly and cost-efficiently examine vast chemical landscapes, reducing potential drug candidates from millions to a more manageable number for subsequent analysis. This capability has profound implications for drug discovery, potentially saving significant amounts of time and resources.^{2–4} Despite its considerable advantages, virtual screening also poses challenges and limitations, particularly concerning the accuracy of predictions.

One of the biggest limitations of VS is the scoring functions used to predict binding affinity. These scoring functions are often rough approximations and usually do not accurately predict the true binding energy or correct binding pose.⁵ They

often struggle to balance speed and accuracy, making it difficult to reduce both false positives and negatives in VS applications.⁶ Additionally, in ligand-based virtual screening, structurally similar compounds are often assumed to have similar activity. However, minor structural changes can sometimes lead to significant changes in activity, a phenomenon known as an “activity cliff”, which can dramatically impact predictive accuracy.⁷

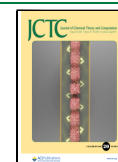
In that context, binding free energy calculation methods offer a unique advantage in the field of computer-aided drug design and virtual screening.⁸ Binding free energy calculations can be more accurate than simple docking scores as they consider not just the static interaction of the ligand and

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protein, but also the dynamic changes that occur upon binding, including conformational changes and solvation effects.⁹ Leveraging binding free energy calculations that rely on physics-based approaches can facilitate the identification of the most promising candidates during virtual screening. Furthermore, they can guide the optimization of lead compounds, indicating the necessary modifications that might improve binding affinity.¹⁰

Methods like Relative Free Energy Perturbation (RFEP) represent the gold standard for predicting binding affinity in the hit-to-lead and lead optimization stages of drug discovery.^{11–13} RFEP is hindered in its application to large libraries due to its high computational cost and the necessity for a structural analog with known activity, which prevents its use in hit identification campaigns.¹⁴ In theory, Absolute FEP (AFEP), which does not require a reference molecule, offers improved accuracy for hit identification.¹⁵ However, in practice, its slow throughput is incompatible with virtual screening, where rapid docking and simple, less reliable, scoring functions remain the norm¹⁶ (see Table 1 for a general speed comparison).

Table 1. General Computational Time Comparison

method	time per ligand on 1 T4 GPU	congeneric ligand required
docking	seconds/mins ^{50,52}	No
MMGBSA	minutes/hours ³⁰	No
AQFEP	1–2h	No
RFEP	1–2 days ⁵³	Yes
AFEP	1 week ¹⁶	No

To overcome the computational cost and low throughput that have made the application of AFEP to virtual screening campaigns infeasible, we make two advancements. First, we present AQFEP, a novel physics-based function based on a fast absolute free energy perturbation method that shows superior ranking performance when compared to other standard scoring functions. This advancement allows us to screen tens of thousands of molecules in a fraction of the time compared to other absolute free energy methods. Second, we combine this increased *in-silico* screening throughput with Bayesian optimization algorithms, to substantially decrease the computational cost of screening the majority of top-scoring compounds with AQFEP. We analyze various surrogate model architectures and acquisition functions to evaluate their effectiveness in efficiently screening the most promising ligands with AQFEP in the context of a virtual screen for drug discovery. We perform these analyses on both straightforward and challenging systems to demonstrate the flexibility of our approach. Finally, we utilize this workflow on a prospective virtual screen through a 1.17 million compound library for hit-finding on a novel protein target.

METHODS

System Preparation and Molecular Docking. The published¹⁷ 3D coordinates of the cMet protein structure and congeneric ligand set were downloaded from <https://github.com/MCompChem/fep-benchmark/tree/master/cmet>. The 3D structure of GLP1R was downloaded from the Protein Data Bank¹⁸ (pdb ID 7S15), and hydrogens were added using the open-source PyMol software and visually inspected. The GLP1R agonist ligands considered in this study,¹⁹ were drawn using JSME²⁰ to generate the SMILES. Using RDKit²¹ in

Knime²² the 3D structures were created, hydrogens added, and protonation states were generated using in-house rule-based SMIRKS and visually inspected.

The commercially available compound library used in the prospective screen was generated starting from the MCULE in stock database <https://mcule.com/database/>, including more than 5 M ligands. The library was deduplicated and filtered in Knime to ensure drug-like properties, removing reactive groups and unwanted chemical moieties. The resulting compounds were visually inspected and further curated using Data-warrior.²³ We focused in particular our analysis on ligands at the extreme of the selected range for the different properties. The final set including 1.17 million ligands was prepared for ligand docking as described above. Part of this library was used for the cMet and GLP1R VS tests. GNINA 1.0²⁴ with the Vinardo²⁵ scoring function was used for all molecular docking studies.

Physics-Based Calculations on Benchmark Data Sets.

For both the cMet and GLP1R benchmarks, poses scored by AQFEP were obtained via a maximum common substructure (MCS) alignment to the pose from the crystal structure found in the PDB (where applicable) and several top poses identified by GNINA using the Vinardo scoring function. Ranking statistics for the different methods and corresponding confidence intervals were generated using the numpy and scipy libraries in Python. 95% confidence intervals were generated via a bootstrap resampling procedure with replacement and 10,000 resampling events. All data generated to perform these benchmarks, including 3D poses and protein structure files, are provided. Details can be found in the Reproducibility and Data Availability Sections.

AQFEP. Below we describe AQFEP whose name comes from a hybrid between Absolute Free Energy Perturbation (AFEP) and SandboxAQ. AQFEP is a physics-based approach to predicting ligand binding affinities that is superior in accuracy to traditional docking scoring methods. Because it is based on first-principles statistical mechanics, AQFEP can handle a wide range of situations (e.g., various ligand charge states, imperfect protein preparation, unusual binding geometries) that would be difficult for traditional scoring functions that are based on fitting to a limited benchmark data set. In addition, compared to traditional physics-based methods, AQFEP is significantly faster. For example, due to the short simulation time (described in more detail below), our approach is not impacted by potential long-term instability related to missing protein regions.

AQFEP uses an absolute free energy perturbation calculation based on the double-decoupling alchemical protocol (Figure 1). The double-decoupling approach is considered the “gold standard” for absolute free energy calculations by ensuring thermodynamic consistency, accurate sampling of the free-energy landscape, and wide applicability to a variety of different systems. For a more detailed description of this method and its advantages, we refer the reader to a comprehensive text on the topic.^{26–28} By using an absolute free energy calculation, which directly calculates the binding free energy of the given ligand-protein pair, the procedure requires much less human guidance than the more typical relative free energy calculations. For example, RFEP calculates differences in binding free energies between pairs of ligands and thus requires careful selection of congeneric ligand pairs and specification of the scaffold relationship between the ligands. While absolute free energy calculations are known for

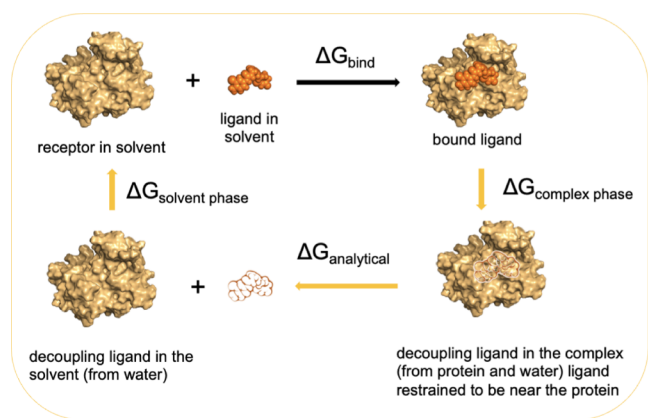


Figure 1. Double-decoupling alchemical protocol.

being difficult to converge and giving inconsistent results unless run for very long simulation times,¹⁵ the AQFEP method is tuned to limit simulation noise and allow for significantly reduced simulation time.

In order to achieve the speeds needed for free energy simulations to be applied to VS workflows we made several adjustments to AQFEP. For example, the simulation time per lambda window was chosen to be shorter than standard free energy perturbation calculations (usually 5 ns¹⁵) in order to evaluate the energy minimum closer to the provided complex conformation. For the method to perform as a scoring function for a given ligand pose, it must ensure it is evaluating a thermodynamic state indicative of that pose. Thus, we have designed AQFEP to sample only the thermodynamic states indicative of that pose during the FEP sampling. In spirit, this is quite different from attempting to sample the full state space in the hope of calculating the partition function and measuring the binding free energy of the ligand-protein pair, which would ignore the given pose of the ligand. In practice, sufficiently sampling the state space for such a calculation requires far too much simulation time, and running the simulations longer than is done in AQFEP only allows the ligand to briefly sample conformations too dissimilar from the proposed pose, significantly increasing the statistical noise. Due to this design choice, the method is very dependent on the quality of the proposed ligand pose(s) which can impact ranking performance (cf. Section System Preparation and Molecular Docking). To ensure the highest quality results, we calculate simulation convergence error for AQFEP using the Multistate Bennett Acceptance Ratio (MBAR) method.²⁹ Although this method is known to underestimate uncertainties, we found in our testing that a cutoff of 1 kcal/mol functions well to separate accurate data from unreliable calculations. A fuller analysis of convergence (e.g., by using a lambda overlap matrix) would likely provide a more accurate assessment of reliability, but is harder to do at the scale of simulations performed in this work. Despite this, AQFEP performs a rigorous, multistep alchemical transformation, unlike end-point methods such as MMGBSA or MMPBSA,³⁰ and is thus significantly more accurate and physically realistic than such end-point methods. For all AQFEP simulations conducted in this paper, the OpenMM MD package was used.

Chemical Space Search Strategy. The goal of virtual screening campaigns is to find some set of high-performing compounds within a large chemical search space. This problem is often solved by exhaustively searching a chemical library to

find the best-scoring compounds according to some scoring function (e.g., molecular docking or AFEP). However, as virtual libraries grow this approach has become computationally infeasible. For example, several recent high-throughput docking campaigns applied to libraries in the hundreds of millions required upward of 40 000 core hours.³ Applying RFEP simulations to libraries that are as small as 10 000 molecules can take upward of 80 000 GPU hours.³¹ AFEP methods are 4–10 times slower than RFEP,¹⁵ resulting in 320,000–800,000 GPU hours (36.5–91.3 GPU years) for 10 000 ligands.

Bayesian optimization^{32–34} is a subset of active learning that helps guide the choice of experiments based on some surrogate models' predictions. Formally, we seek to find the set of top- k molecules (M) from a chemical library ($\mathcal{D}$) that maximizes some black-box objective function (here AQFEP of a candidate compound):

Top- k scoring molecules $x \in M$ where M is such that

$$\operatorname{argmax}_{M \subset \mathcal{D}: |M|=k} \sum_{x \in M} f(x) \quad (1)$$

We begin by first calculating the objective function $f(x)$ on a set of n points. The evaluations of this function are stored in the data set L which contains the labeled observations (AQFEP score for a given candidate compound). A surrogate model $\hat{f}(x)$ is trained with this data set and makes predictions on the remaining unlabeled set of data D' . The model predictions are passed to an acquisition function α that determines the utility of acquiring new data points to be labeled. These selected points are then evaluated and added back into the labeled set L . This process is repeated until some stopping criteria are met. In this work, we use a fixed budget size divided into T steps to determine when the algorithm (Algorithm 1) should stop acquiring labels.

Box 1. Algorithm 1 Bayesian Optimization

Input: objective function $f(x)$, acquisition function α , surrogate model $\hat{f}(x)$, and some chemical library D

Select random batch $S \subset D$

Evaluate objective $f(x)$ to generate labels y_s for $s \in S$

Initialize L , the labeled set of data (s, Y_s) for $s \in S$

for $t \leftarrow 1$ to T **do**

Train surrogate model $\hat{f}(x)$ using labeled data set L

Select new batch $S_t \subset D$ using acquisition function α

Evaluate objective function $f(x)$ on S_t

Update L with new labeled batch

end for

Acquisition Function. Below we provide brief descriptions of the acquisition functions used in this work. We refer the reader to other works that describe these acquisition functions in more detail.³⁵

In this work, we evaluated the following acquisition functions:

$$\text{Random}(x) \sim U(0, 1) \quad (2)$$

$$\text{Greedy}(x) = \hat{\mu}(x) \quad (3)$$

Greedy is a naive method where the top-scoring molecules ranked by the model are always selected for evaluation, without taking into account uncertainty.

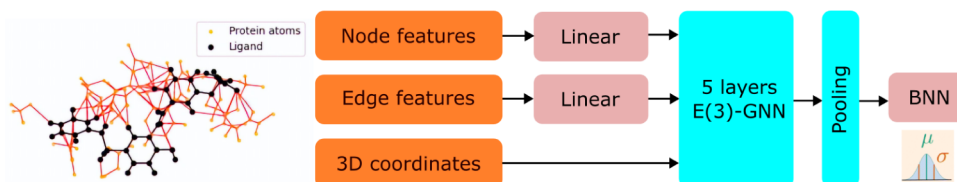


Figure 2. GraphDock architecture. EGNN takes 3 inputs: positions, node embeddings, and edge embeddings. If there are n atoms in an input structure, the position tensor is an $n \times 3$ tensor containing the x , y , and z coordinates of each atom. Edges in the protein–ligand graph are generated using a 3 Å cutoff distance. Nodes that fall outside of this 3 Å radius are ignored. The protein–ligand graph is then further truncated to include only atoms within 3 nearest-neighbor connections to the ligand atoms in order to reduce the complexity of the graph.

$$\text{Epsilon Greedy}(x) = \begin{cases} \hat{\mu}(x), & \text{with probability } 1 - \epsilon \\ U(0, 1), & \text{with probability } \epsilon \end{cases} \quad (4)$$

Epsilon greedy is analogous to the greedy strategy except that it makes a random selection from the library with probability ϵ . In experiments we report, a value of $\epsilon = 0.1$ was used.

$$\text{UCB}(x) = \widehat{\mu}(x) + \beta \hat{\sigma}(x) \quad (5)$$

The upper confidence bound acquisition policy is an “optimistic” method that selects molecules based on their potential to yield optimal values. The utility of acquiring a given molecule is calculated by summing the predicted mean value with its predicted standard deviation. It attempts to balance exploration and exploitation by enabling molecules to be selected that have moderate mean predicted scores but large standard deviations. The β parameter can be adjusted to more heavily weight the standard deviation term. In the experiments we report, a value of $\beta = 2$ was used.

$$\text{PI}(x) = \begin{cases} \Phi(z), & \hat{\sigma}(x) > 0 \\ 1, & \hat{\sigma}(x) = 0 \text{ and } \gamma(x) > 0 \\ 0, & \hat{\sigma}(x) = 0 \text{ and } \gamma(x) \leq 0 \end{cases} \quad (6)$$

The probability of improvement (PI) acquisition policy aims to select the molecule that has the highest probability of improving upon the currently identified best score. The PI score for a molecule is computed utilizing the standard deviation associated with that molecule to compute the amount of probability mass that molecule has above the current best solution. It is important to note that PI does not consider the magnitude of the improvement.

$$\text{EI}(x) = \begin{cases} \gamma(x)\Phi(z) + \hat{\sigma}(x)\phi(z), & \hat{\sigma} > 0 \\ \gamma(x), & \hat{\sigma} = 0 \end{cases} \quad (7)$$

The expected improvement policy (EI) is analogous to the PI policy but considers the magnitude of the improvement. The value is computed for a molecule by calculating the expected value of the probability density that the molecule has above the current best solution. The EI method can be augmented with a parameter that can encourage more exploration.

For PI and EI:
 $\gamma(x) := \hat{\mu}(x) - f^* + \xi$; $z(x) := \frac{\gamma(x)}{\hat{\sigma}(x)}$; $\hat{\mu}(x)$ and $\hat{\sigma}^2$ are the surrogate models predicted mean and uncertainties for point x , respectively. Φ and ϕ are the CDF and PDF of the standard normal distribution and f^* is the current maximum

objective function value. In experiments we report using EI and PI, we use a value of $\xi = 0.01$.

Surrogate Models. Random Forest. Random forest regression is an ensemble learning technique that utilizes a set of decision trees. Each individual tree is fit with a random subset of the training features and observations in an attempt to decorrelate the trees.^{36–38} During inference, uncertainty estimates can be derived by examining the mean value of the ensemble of trees and the variance of the predictions from the ensemble. Chemical libraries were featurized with molecular fingerprints. There are a variety of molecular fingerprints that differ in their specific implementations, however, they all can be broadly understood as representing the presence or absence of a specific substructure within a molecule into a vector of fixed length.³⁹ In this work, we utilized the Morgan Fingerprint with a bit length of 2048 and a radius of 3. The RF surrogate used was fit with $n_{\text{estimators}} = 100$, $\text{max_depth} = 8$.

D-MPNN. In this work, we utilized the directed message-passing neural network (D-MPNN) implemented by.⁴⁰ MPNNs treat the molecule as a connected graph and construct a feature vector for that graph, denovo.⁴¹ This is in contrast to fixed fingerprints which do not have the flexibility to adjust their embeddings. Broadly, MPNNs operate in two stages, the message-passing phase and the readout phase. In the message-passing phase, “messages” are passed between atoms and/or bonds and their direct neighbors, and incoming messages are used to update the “hidden state” of each atom and/or bond. The message passing phase is repeated over multiple (e.g., 3) iterations, at which point the hidden states of each atom are aggregated (e.g., summed) to produce a molecule-level feature vector. By training this model at the same time as an FFNN operating on the feature vector, MPNNs are able to learn a task-specific representation of an input molecular graph. For more details on the D-MPNN model and the specific implementation of this architecture, we refer the reader to.⁴⁰

The message-passing neural network used here utilized standard settings from the molecule model class in the Chemprop library:⁴⁰ messages passed on directed bonds, messages subjected to a rectified linear unit (ReLU) activation, a learned encoded representation of dimension 300, and the output of the message passing phase fully connected to an output layer of size 1. The model was trained using the Adam optimization algorithm, a Noam learning rate scheduler (initial, maximum, and final learning rates of 0.1, 0.001, and 0.0001, respectively), and a root-mean-squared error loss function over 50 epochs with a batch size of 50. For more details on the Noam learning rate scheduler, see.⁴² The model was trained with early stopping tracking the validation score using a patience value of 10. When uncertainty values were needed for metric function calculation, a mean variance estimator (MVE) model based on the work done by⁴³ was

used. This model featured an output size of two and was trained using the loss function defined by Nix and Weigend.⁴⁴

GraphDock. The D-MPNN model presented in the previous section fundamentally utilizes only 2D information about the molecule in question. Moreover, by design, it does not utilize information about the 3D conformation of the molecule, nor the interactions of the molecule with the protein. In this work, we implement a 3D graph neural network, similar to the PointVS model described in ref 45 (Figure 2). This model is an E(n) equivariant graph neural network model, that operates on the 3D protein–ligand complex. The E(n) GNN has demonstrated state-of-the-art performance in regression and classification of chemical data sets⁴⁶ while avoiding the use of computationally expensive spherical harmonics. The model is capable of exploiting the symmetries in the protein–ligand complex without the need for augmenting the data set with translations, rotations, and reflections, as is required for nonequivariant networks such as 3D convolutional neural networks.⁴⁷

We make several adjustments to our model to make it more effective for small data sets and easier to use in Bayesian optimization algorithms where prediction uncertainties are needed. Specifically, GraphDock is composed of 5 layers of E(3)GNN, with 100 hidden dimensions, and the same parameters for chemical prediction tasks specified in.⁴⁶ Node and edge representations first pass through separate feed-forward neural networks, each with hidden dimensions 32. The output graph is pooled to give an embedding with 64 dimensions (see Figure 2).

In the PointVS paper, the authors use 48 EGNN layers, we reduce the number to 5 layers in this work to reduce computational complexity and improve training time.⁴⁵ In addition, we append a single Bayesian linear layer, with a Gaussian prior, for use in regression tasks and in order to save computational costs associated with estimating uncertainty from the model.^{48,49} Protein–ligand graphs were constructed using the same protein and ligand conformation as used for AQFEP scoring. In addition, edges in the protein–ligand graph were generated using a 3 Å cutoff distance. This cutoff was selected because it approximates intermolecular interactions such as hydrogen bonding within the complex. The connected protein–ligand complex is then truncated to only atoms within 3 nearest-neighbor connections to the ligand atoms. The graph featurization step utilizes the same feature set as used in the Chemprop library,⁴⁰ in addition to the 3D coordinates, and the identity of a node (atom in protein or in ligand). This was done to ensure that the addition of features related to the protein and 3D information improved model performance. The GraphDock model was trained using the Adam optimization algorithm with a batch size of 64 and a fixed learning rate of 5×10^{-4} for 500 epochs. Early stopping was used to prevent overfitting with a patience of 50.

Comparison to Other Methods. To evaluate different surrogate models' performance, we compared two commonly used search strategies: random search and top dockers. In this case, random search is indicative of an exhaustive search strategy where every compound has an equal probability of being evaluated. In addition, we also evaluated the performance of selecting compounds based on their docking scores (Top Dockers) obtained from the GNINA docking software⁵⁰ with the Vinardo scoring function.²⁵ This is a useful comparison because many VS campaigns use the top-docking compounds from a primary screen to select which compounds

to empirically validate or perform additional high-fidelity simulations on.³ Due to computational limitations, we did not redock the compounds with different random seeds and in the figures reported in the paper, there are no error bars reported for this condition.

Evaluation Metrics. The active learning strategy was evaluated in several different ways. For prospective searches, we use the median top-k AQFEP score identified (lower is better), as well as the number of molecules identified that have AQFEP scores below -20 kcal/mol. For retrospective searches, we use the fraction of top-k SMILES identified in a precomputed library and the number of times a specific active-learning workflow found a known active compound (hit rate).

Reproducibility. Where possible we have shared 3D poses, SMILES, prepared protein files, details of molecular docking approaches, and scores for all experiments performed in the paper. However, the full AQFEP workflow is proprietary and comprehensive technical details have not been provided in this manuscript. All methodological and technical details of both the machine learning approaches and bayesian optimization approaches have been shared in full in this manuscript.

RESULTS AND DISCUSSION

AQFEP Ranking Performance and Speed Comparison. In order to evaluate the effectiveness of AQFEP applied to VS workflows, we tested the accuracy of AQFEP on the cMet protein kinase using two common applications of interest for drug discovery: 1) evaluation of the free energy of binding on a set of congeneric ligands; 2) application to virtual screening. In addition, we also performed general speed comparisons between commonly used methods in structure-based virtual screening.

Speed Comparison. To increase the ligand throughput of AQFEP so that it could be used within a VS workflow, several changes were implemented in AQFEP. Most notably, the MD simulation time for each lambda window was chosen to be shorter than standard free energy perturbation calculations. This adjustment allows AQFEP to be used at a throughput much higher than traditional free energy perturbation methods, allowing the virtual screening of libraries of tens of thousands of ligands. For instance, using 20k NVIDIA T4 GPUs, AQFEP can reach 10–20k ligands profiled per hour (Table 1). However, this design choice causes the method to be very dependent on the quality of the proposed ligand pose(s) which can impact ranking performance. In the sections that follow, we highlight this effect and show its impact on downstream scoring with AQFEP.

Congeneric Ligand Test. RFEP is the state-of-the-art approach for evaluating the difference in free energy of binding of related ligands, and a large number of benchmark sets are available in the literature.¹² It is commonly used in industry in hit-to-lead and lead-optimization. RFEP is known to be challenging for large perturbations, changes affecting opposite regions of the molecule simultaneously, charge perturbations, large scaffold changes, and perturbations affecting a linker region in the molecule.⁵⁴ AQFEP can be used in all these cases, is much faster to compute (Table 2), and it does not require an expert user to carefully superimpose the ligand to the reference compound. When evaluating performance on the cMet benchmark set,¹⁷ we find promising ranking performance (Figure 3), that is close to FEP+, but at a fraction of the computational cost. The results show clear

Table 2. Ranking Performance of the 5 Active Ligands for AQFEP vs Docking using Vinardo

ligand	IC ₅₀ (μ M)	Vinardo rank	AQFEP rank
CHEMBL3402751	2.1	550	13
CHEMBL3402747	3.4	53	20
CHEMBL3402755	4.2	9	7
CHEMBL3402748	5.3	124	9
CHEMBL3402752	30	870	49

improvements compared to MMGBSA, or molecular docking (Glide and GNINA/Vinardo).

Since AQFEP's MD simulation time is limited for each lambda window to maximize speed, the final prediction will be strongly influenced by the energy minimum closest to the starting conformation. If the starting conformation is far from the physically most important energy well, it is questionable if MD, even with enhanced sampling, will be the best approach to identify the correct lowest energy configuration. To quantify the impact of incorrect ligand conformation on downstream affinity predictions using AQFEP, we rescored the top 10 poses generated from GNINA/Vinardo in the cMet benchmark described above. Rescoring with AQFEP the second and third top poses from GNINA/Vinardo (Figure 3) improved the ranking performance of our method. Considering additional poses beyond 3 did not result in large changes in the ranking performance of the method. For this particular case, using the Vinardo top-scoring pose for each ligand in AQFEP resulted in better performance compared to rescoring ligand positions

generated from ligand alignment (shown as pose 0 in the plot in Figure 3).

VS Enrichment Test. We created a realistic VS test set including 1 active ligand for every 54,000 commercially available drug-like compounds. The 5 low micromolar inhibitors of cMet were selected from the previously discussed congeneric series test set.¹⁷ For this evaluation we used a classical funnel approach, selecting the top 1000 ligands from molecular docking for AQFEP rescoring. Protein kinases are known to be ideal cases for molecular docking and the Vinardo scoring function was able to rank the 5 ligands in the top 1k compounds (Figure 4). Rescoring with AQFEP identified 2 actives among the top 10 compounds and all 5 active ligands in the top 50 (Table 2). This analysis showed a clear advantage for AQFEP in early enrichment ranking performance when compared to Vinardo.

ML Driven Chemical Space Exploration Maximizes Efficiency and Hit Rate. Retrospective Test—GLP1R. Despite large improvements in computational efficiency, AQFEP in practice is still too slow to be applied to the ultralarge virtual screens that have now become the norm. For example, ZINC, a popular open-source database of commercially available compounds, now has close to 30 billion entries.⁵⁵ These libraries are challenging to screen against even with standard structure-based drug design tools (e.g., docking) and were previously thought to be unattainable for free energy calculations. Search optimization strategies exist to efficiently search through chemical libraries for compounds that have desirable properties, such as active learning or Bayesian

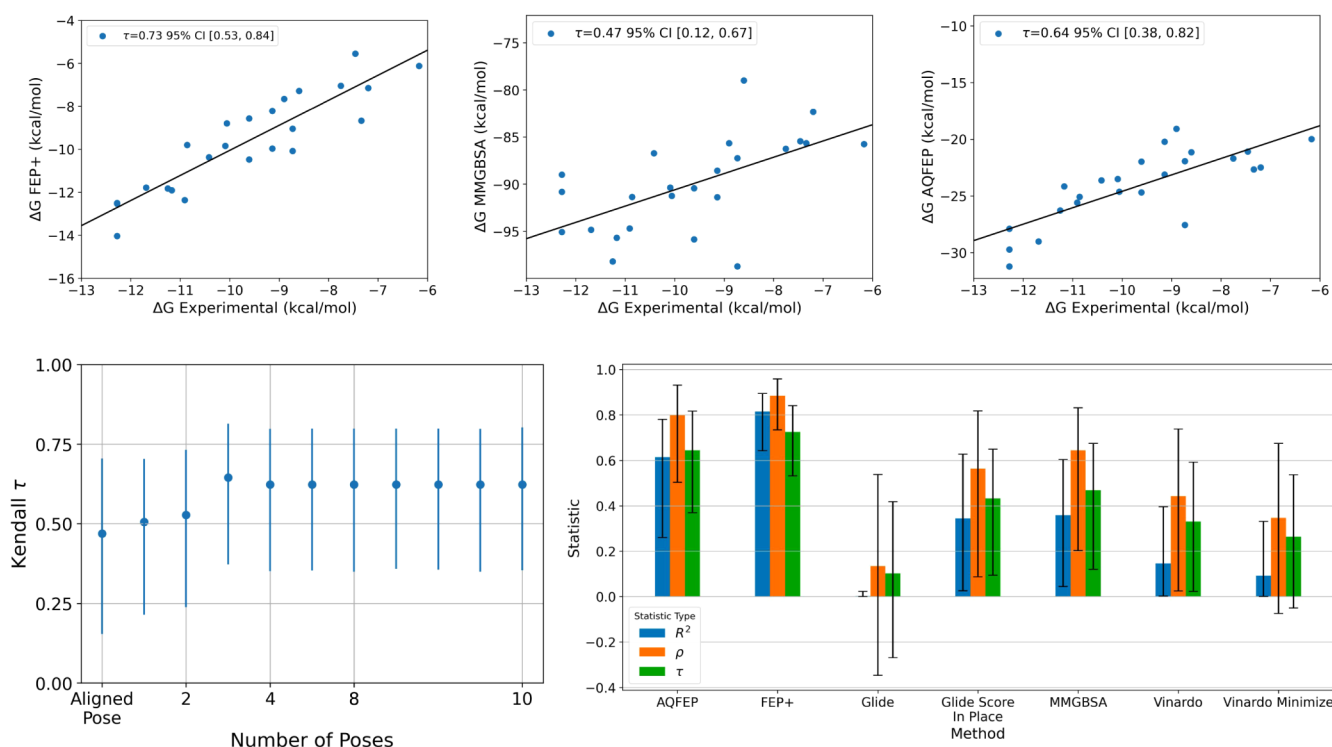


Figure 3. Comparison of the predicted vs experimental free energy of binding for FEP+¹⁷ (top left), Prime MMGBSA¹⁷ (top center), and AQFEP (top right). The effect of binding pose prediction on AQFEP is shown in the bottom left plot. The X-axis includes the number of top poses from GNINA/Vinardo scoring function profiled with AQFEP for each ligand. On the Y-axis is the reported Kendall tau rank-correlation based on the best AQFEP scoring poses. Aligned Pose corresponds to the ligand position from simple ligand alignment available from ref 17. The bottom right plot includes the comparison of the correlation and ranking performance of AQFEP compared to other commonly used methods. Glide $R^2 = 0.002$ is close to the X-axis and therefore not visible in the plot. 95% confidence intervals for plots in the bottom row were estimated with bootstrap resampling with replacement and 10,000 resampling iterations.

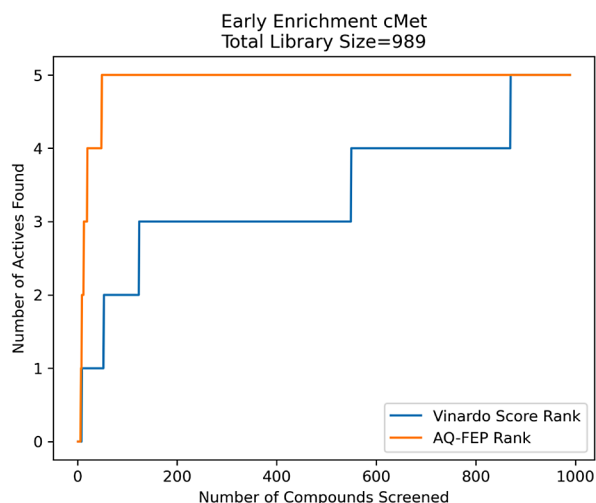


Figure 4. Ranking performance on cMet enrichment test set with 5 known actives. Enrichment using a standard docking workflow performs well and is improved further using AQFEP.

Optimization (BO). However, due to the computational cost of free energy methods, generating sufficiently large quantities of data needed to train models that are broadly generalizable to vast chemical space is not possible. Bayesian optimization and active learning have been previously applied to virtual screening campaigns using molecular docking, e.g., refs ^{56,57} and RFEP,^{14,31} but to our knowledge, we disclose its application to the largest set of AFEP calculations to date.

To mitigate the impact of growing library size, we employ a Bayesian optimization strategy to perform a model-guided search and seek to find a set of top-*k* molecules that optimize some criteria with the fewest number of black-box function calls. In this study, the black box function is AQFEP and we seek to find a set of top-*k* molecules that have the lowest binding affinity as measured with this function.

During each iteration of optimization, newly evaluated compounds are appended to the training set and used to train a surrogate model. Here, we define our surrogate as some supervised machine learning model trained with labels generated from AQFEP simulations and use these surrogate model's predictions to guide the acquisition of AQFEP labels. The mean and variance of the model outputs are used to calculate the acquisition function and define the utility of

acquiring labels for compounds in the next batch (see Methods—Chemical Space Search Strategy).

We tested this approach on the GLP1R receptor, a challenging Family B GPCR membrane protein related to type 2 diabetes.⁵⁸ We included two weak agonists as actives¹⁹ in a library of more than 12,000 commercially available ligands with drug-like properties. As an initial evaluation to demonstrate how AQFEP and Bayesian optimization algorithms can be used together, we generated a data set of 12,720 AQFEP scores screened against GLP1R. As mentioned previously, we also included in this library 2 known active molecules to evaluate the ability of our method to screen known actives in the presence of a large number of decoys. Data acquisition was simulated with an initial random 2% selection followed by sequential 2% acquisitions for 5 iterations totaling a 12% total library screen. We test several surrogate models that are typically used in Bayesian optimization strategies including Random Forest (RF) with Morgan fingerprints and D-MPNN.⁴⁰ In addition, we also compare these to a 3D Graph Neural Network architecture (GraphDock) (2). To quantify the advantages of model-guided search, we benchmarked our approach against random search, indicative of an exhaustive search through the library, and the top-scoring compounds from the docking program GNINA using the Vinardo scoring function (see Methods).

Bayesian optimization using any of the surrogate models tested, yields clear improvements over other heuristic search algorithms. Random Forest operating on molecular fingerprints showed similar performance across acquisition functions in the top-*k* task with EI, Greedy, and ϵ -greedy all performing similarly (Figure 5 and Table 3). The greedy strategy performs best, which finds (253.2 ± 13.5) of the top-performing compounds. A similar pattern emerges when examining the D-MPNN surrogate with both greedy strategies performing best (Greedy: 267.5 ± 5.0 ; ϵ -greedy: 259.6 ± 11.2). The GraphDock model performed the best in the top-*k* task where greedy strategies perform best, with UCB finding 322.0 ± 7.3 of the top-*k*. Surprisingly, docking performs only marginally better than random search on the GLP1R benchmark, with top dockers finding 73 and random finding 58.6 ± 5.95 of the top-*k* in this library. We note that past work in this area,^{31,57} investigated the impact of initial seed size on Bayesian optimization performance. They found that increasing this value significantly improved performance in the top-*k*. We have not performed those experiments, but expect

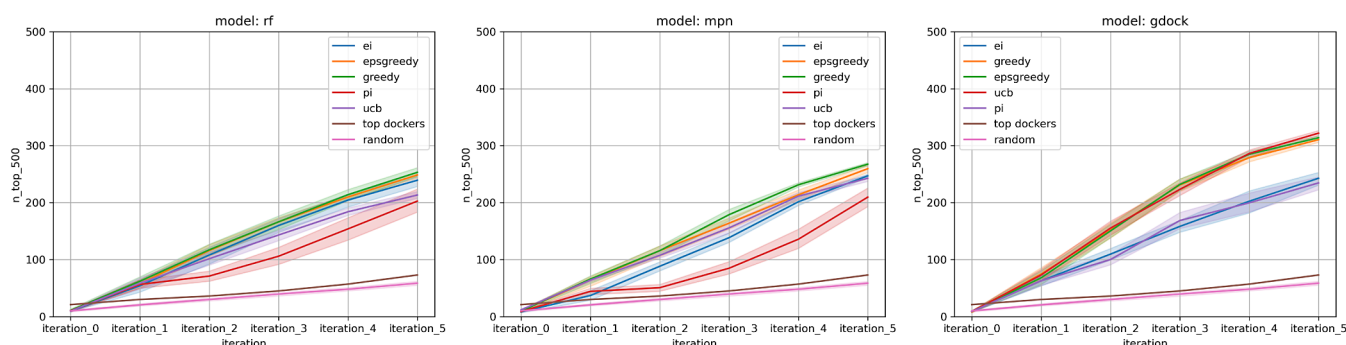


Figure 5. Bayesian optimization performance on GLP1R retrospective test as measured by the number of top 500 scores found as a function of total compounds evaluated. Each trace represents the performance of a given surrogate with a specific acquisition function. Each experiment began with 2% random selection followed by 5 iterations of 2%, for a total of 12% (1530 compounds). The total library size was 12,720. Error bars reflect $\pm$ one standard deviation across 10 runs.

Table 3. Average Top-k Performance on GLP1R Retrospective Test with a 2% Acquisition Size as Measured by the Average Number (n) of Top Smiles Evaluated in the Pre-Computed Library Across 10 Experiment Runs^a

surrogate	acq	mean	std
gdock	ei	242.8	19.7
	epsgreedy	314.0	5.9
	greedy	310.6	8.2
	pi	234.6	18.5
	ucb	322.0	7.3
mpn	ei	247.0	5.0
	epsgreedy	259.6	11.2
	greedy	267.5	5.0
	pi	209.7	27.4
	ucb	242.7	10.0
rf	ei	239.3	16.9
	epsgreedy	248.6	9.8
	greedy	253.2	13.5
	pi	202.7	35.0
	ucb	213.4	10.1
top dockers	top dockers	73.0	NA
random	random	58.6	5.9

^aBolded numbers indicate the best-performing method within each surrogate model.

adjusting this value to have a similar impact on the work reported here.

An alternative way of understanding the performance of the surrogate model is to calculate the number of times each screening workflow identified one of the known actives placed within this library (Figure 6 and Table 4). The random forest model utilizing Morgan fingerprints performed remarkably well with the ϵ -greedy acquisition function having an average hit rate of 0.12 ± 0.04 . When using the D-MPNN as a surrogate, two different greedy acquisition functions were found both known actives in 10/10 runs (max hit rate is 0.13% 2/1530). Interestingly, while GraphDock performed quite well in the top-k task, this model's hit rate was low when compared to other surrogates, with the greedy acquisition function having an average hit rate of 0.08 ± 0.03 . This effect may be due to the structural similarity of the two active compounds helping a 2D approach (D-MPNN and RF) to identify both simultaneously. In contrast, GraphDock, is affected by the quality of

Table 4. Average Hit-Rate Performance on GLP1R Retrospective Test

surrogate	acq	mean	std
theoretical max	NA	0.130	NA
gdock	ei	0.013	0.028
	epsgreedy	0.072	0.021
	greedy	0.079	0.028
	pi	0.020	0.032
	ucb	0.060	0.021
mpn	ei	0.124	0.021
	epsgreedy	0.124	0.021
	greedy	0.130	0.000
	pi	0.098	0.046
	ucb	0.130	0.000
rf	ei	0.111	0.044
	epsgreedy	0.118	0.041
	greedy	0.092	0.063
	pi	0.072	0.057
	ucb	0.107	0.044
top dockers	top dockers	0.000	NA
random	random	0.000	0.000

the docked poses used to train the model. To further investigate, we examined the individual poses of the known actives that were generated with GNINA/Vinardo. We found that one of the two active compounds had a predicted pose that was quite different from the X-ray conformation of related agonist compounds. In addition, AQFEP scores for the active compounds in this library had relatively low ranks and both active compounds did not place in the top 500 in the precomputed library of 12,720 (Active 1 FEP Rank = 1102, Active 2 FEP Rank = 1737). It is likely that this wrong pose, combined with relatively low AQFEP scores for the actives, hampered the ability of GraphDock to identify this active ligand. Finally, both random search and docking fail to screen any known active ligands.

Prospective Test—Novel Protein Target. We next turned to evaluate the effectiveness of this workflow applied to a prospective search in a much larger chemical space. Here, we screened a 1.17 m compound library of commercially available compounds (MCULE) against a novel protein target. The protein target used in this prospective study has no known active chemical matter reported in the literature and for all

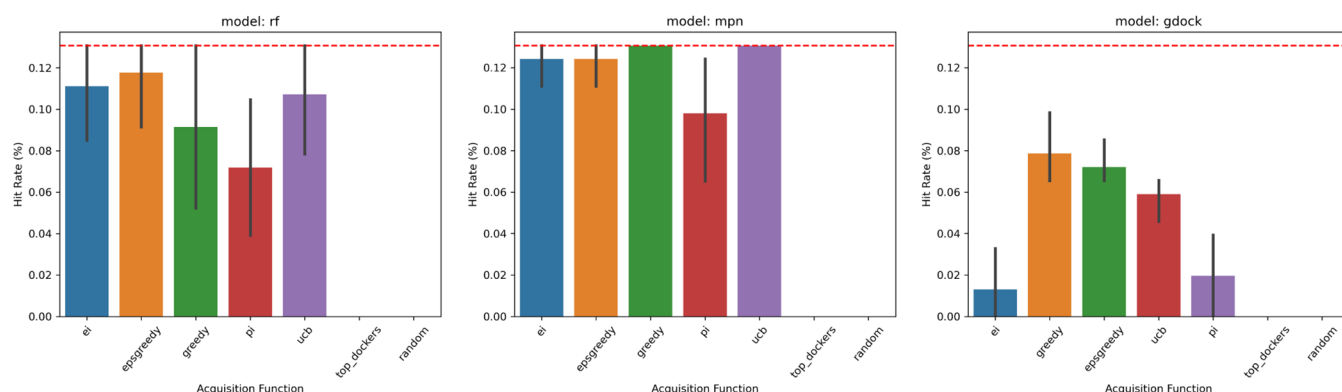


Figure 6. Bayesian optimization performance on GLP1R retrospective test as measured by the average hit rate across 10 runs. Each bar represents the performance of a given surrogate with a specific acquisition function and indicates how often a given surrogate-acquisition pairing can find both known actives in the library. The red dashed line indicates the theoretical max performance (e.g., finding both actives in all 10 runs). The red dashed line is 0.13% (2/1530). Error bars reflect $\pm$ one standard deviation across 10 runs.

intent and purposes is considered undruggable. In this experiment, we performed two internal controls to show that our method was intelligently sampling the chemical space. First, we randomly selected 10,000 compounds to evaluate the number of compounds that have low AQFEP scores. Second, we docked the entire library (1.17m) with GNINA and selected 10,000 compounds with the lowest (best) docking scores. From here we can establish a baseline on how conventional heuristic selection strategies score with AQFEP. To ensure the highest quality labels for our algorithm, we removed AQFEP scores from this set if the simulation had an uncertainty greater than 1 kcal/mol (as estimated by the multistate Bennett acceptance ratio).²⁹ This resulted in 7,542 and 8,930 ligands in the Random and Top Dockers selection, respectively.

We used this heterogeneous sample (random selection plus top-dockers) to serve as our initial seed set of labeled data in our active learning workflow. We performed 3 iterations of active learning with a 10,000 compound acquisition size. In each iteration, we also examined the convergence error for our AQFEP scores, and scores with greater than 1 kcal/mol error were removed. This resulted in 6,792, 3,317, and 3,527 compounds profiled in tranches 3 through 5 respectively. In total, we performed 30,108 free energy simulations throughout our active learning workflow, representing the largest reported free energy screens to date. We can understand how well our active learning strategy is performing by looking for changes in the median AQ-FEP score across iterations. If our algorithm is performing as intended, we should see the median AQ-FEP score decrease across active learning iterations. Utilizing a D-MPNN surrogate, with a greedy acquisition function, we can see in Figure 7, that across iterations, model-guided selection

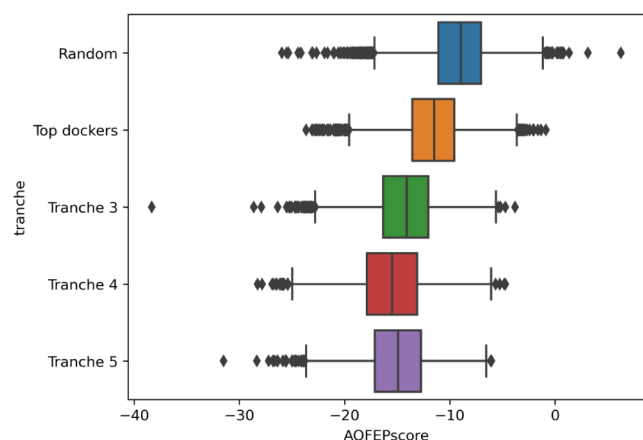


Figure 7. Model guided search results in profiling of compounds with better (lower) AQ-FEP scores across tranches against a novel protein target. In total 30,108 compounds were profiled which is comprised of 7,542 compounds from Random selection, 8,930 from Top Dockers selection, and 13,636 selected by a greedy D-MPNN surrogate model.

selects consistently lower-scoring compounds across iterations, and is substantially better than both internal controls. Based on the analysis of the AQFEP free energy of binding distribution for the random sample we defined a promising ligand to have a predicted score of -20 kcal/mol. The number of such promising ligands identified in each iteration (Table 5) showcases the efficiency of the proposed method. After completing the active learning iterations, 1424 of the top scoring compounds (AQFEP Score) were selected and

Table 5. Number of Ligands with Low AQ-FEP Scores Across Tranches

search strategy	number of ligands identified with AQFEP < -20 kcal/mol
random	16
top dockers	46
Tranche 3 (D-MPNN)	305
Tranche 4 (D-MPNN)	351
Tranche 5 (D-MPNN)	267

purchased for further experimental validation using a primary binding assay. Virtual screening hits were defined as having potency values of ≤ 50 μm . 41 diverse compounds were found to be active using these criteria, which represents a primary hit-rate of $\sim 2.88\%$.

CONCLUSION

In this work, we describe AQFEP, a physics-based approach for evaluating the free energy of binding for diverse ligands to protein targets that is superior to molecular docking scoring functions in predicting ligand binding free energy. The speed of AQFEP enables the virtual screening of libraries of tens of thousands of ligands at unprecedented accuracy in a target-independent way. Compared to RFEP, AQFEP does not require a similar compound with known activity, it can be used to score a pose directly after docking without the need for careful alignment to a congeneric compound. It is significantly faster, with one ligand typically taking 1–2 h on one Nvidia T4 GPU, approximately 10 times faster than RFEP and 40–70 times faster than other AFEP solutions on comparable hardware.

Most importantly, the speed of AQFEP also enables the generation of high-quality labeled data sets large enough for the training of a variety of supervised machine learning models. These models achieve speeds comparable to standard docking scoring functions. A pose-dependent 3D-EGNN using AQFEP labels can be used as a surrogate model in a Bayesian optimization framework enabling efficient search through libraries of millions of ligands. This potentially reduces the need for profiling additional ligands with AQFEP to only those that are structurally novel.

This ML-guided search of chemical space using AQFEP as an objective function shows improvements in both hit rate and percent of top compounds screened when compared to random and other heuristic search algorithms (top dockers). Across Bayesian optimization iterations in a prospective search, a greedy D-MPNN surrogate model selects lower-scoring compounds than other selection strategies (top dockers and random search).

Together, this work demonstrates the effectiveness of the unification of a fast and accurate physics-based scoring function with BO algorithms to unlock the capability to perform large virtual screens using free energy calculations.

ASSOCIATED CONTENT

Data Availability Statement

Raw and processed data to reproduce figures in this manuscript are available in a Zenodo repository.⁵¹

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Notes

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